

Exploration on the Effectiveness of Learning, Interest, and Attitude of the Integration of Review System of History Based on Mobile Game and Forgetting Curve

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Abstract. According to Ebbinghaus' forgetting curve theory, memory is unreliable and a substantial amount of information is lost immediately after learning. Therefore, this study constructed a review system based on mobile gaming, which was provided to students as a review tool within 20 min of completing a history lesson. The aim of this study was to enhance student effectiveness, attitude, and interest. The research tool included three exams and questionnaires regarding learning attitude and learning interest. The learning scores did not exhibit significant differences between the control and the experimental group; however, the mean for the experimental group was greater than that for the control group. Moreover, learning attitude and interest were significantly different between the two groups.

Keywords: Forgetting curve · History · Digital-game based learning

1 Introduction

Courses provided during primary education in a 12-year public education system aim to cultivate students' basic knowledge and abilities while deepening their sense of learning. Students are expected to care about local events and recognize and embrace cultural diversity. Numerous fields of social studies exist, among which history, involving rich changes and influences, explores the origin of culture from an objective perspective [1]. Current studies focus largely on major subjects such as Chinese, English, and mathematics. By contrast, the current study centered on history courses and attempted to emphasize the overall importance of social studies in education. Moreover, this study applied information technology to strengthen students' motivation to study.

Changes in information technology have engendered changes in traditional learning environments. With the rapid development of information technology, the acquisition, application, and communication of information have undergone tremendous changes [2]. Furthermore, information technology is currently widely incorporated into general

learning [3]. Educational environments complemented by information technology are commonly available. The increasing popularity of wireless technology and mobile devices has propelled mobile learning [4].

Among diversified learning environments, digital game-based learning approaches engender enjoyment during learning. Specifically, gaming can enhance enjoyment and enthusiasm during the learning process. Game-based learning attracts learners and generates motivation. As students advance in games, more problem-solving skills are “unlocked.” Students not only achieve learning goals and strengthen their confidence, but also experience an increased desire to learn more and obtain a sense of satisfaction and achievement. Hence, joyful learning leads to positive attitudes. Game-based learning therefore generates more motivation than traditional learning environments [5].

Games are rich in media properties, integrating animation, text, sound, video, and other related content [6]. Game design focuses on a student-centered education module and integrates learning content. The principal goal is to improve student problem-solving abilities [7]. As students pass gateways, they achieve learning goals; deepen their learning impressions; and enhance their interest, effectiveness, and attitude toward learning. Learning attitude refers to enthusiasm, initiative, and a strong recognition of content. Learning attitudes have shifted from passive learning in the past to systems that involve searching for relevant course content. Increased learning interest results from the combination of gaming and social studies courses, which enhances student interest in course content, engenders different forms of media, offers diversified and lively learning environments, and improves previous learning models.

The objective of the current study was to enhance student effectiveness, attitude, and interest toward learning. To study effectiveness, this study adopted the forgetting curve proposed by Hermann Ebbinghaus. This curve asserts that the 20 min after learning is the period wherein people forget the most information. To strengthen student memory, a game-based review system designed by this study was introduced within the 20 min of information input. A tablet was added to the teaching materials. The games generated excitement and stimulation and enhanced learner attitude and interest.

The practical mobile devices can complement the lack of a traditional learning environment, encouraging student confidence and active participation in the learning process [8]. Our system combines gaming and review questions, and implement in the tablet device. The purpose of the system is to provide an alternative to traditional review methods. Student attitude and interest were assessed following their use of our game-based review system.

2 Literature Review

2.1 Digital-Game Based Learning

Game-based learning introduces game and software into a learning environment to help students gain knowledge and enhance ability. Problem solving and challenges within games can provide learners with context and atmosphere to facilitate the achievement of goals [9].

In the game-based learning system, design elements included goals, conditions and rules, outcome, and impact on player [10]. The following conditions must be met to immerse students in the game:

- (1) The game should incorporate appropriate goals and objects, and the subject of each game should be clear.
- (2) Real-time feedback should be provided to enable students to promptly evaluate their learning outcomes as they progress.
- (3) Rules should be coherent so that learners can be compliant to and understand the organization and content of the game, thus enabling them to effectively realize their learning goals.
- (4) Students should progress in a logical, step-by-step manner in the game to enable them adapt to the learning context.
- (5) Interest in learning should be enhanced by combining the game with the course.
- (6) The sound, animation, and visuals of the game should attract attention.
- (7) Student motivation should be maintained.
- (8) The game should be designed to adapt to individual progress, create a competitive and comparative learning atmosphere, and generate a “survival-of-the-fittest” mentality. This can consequently increase their engagement in learning and motivation to promote their interest. Concurrently, the students can focus on the learning context, thus enhancing their attitude toward the social studies.

2.2 Forgetting Curve

The learning process comprises two parts: retention and forgetting [11]. Forgetting rates vary individually. Forgetting refers to a loss of knowledge after the input of information [12]. In this study, the measured forgetting rate was extremely fast. Table 1 shows Ebbinghaus’ forgetting curve, indicating that the forgetting rate slowed over time. Therefore, this study introduced the review system within 20 min of learning to prolong students’ memory and enhance learning effectiveness.

Table 1. Ebbinghaus’ forgetting curve

After learning	Memory retention	Memory left
Learning Just finished	100%	0%
20 min	58%	42%
1 h	44%	56%
9 h	36%	64%
1 day	34%	66%
2 days	28%	72%
6 days	25%	75%
31 days	21%	79%

3 Method

This study investigated the learning outcome, interest, and attitude of students after using a game-based review system following study. In contrast to traditional assessment methods, this study applied a digital game-based review system and Ebbinghaus' forgetting curve, and introduced the system within 20 min after information input, a period during which students had the fastest forgetting rate. Review of the course content helped students prolong their memory and enhance their academic achievement.

3.1 Participant

Two classes of fourth-grade primary school students served as the subjects. A total of 44 students were grouped according to classroom, wherein a class of 21 students constituted the experimental group and another class of 23 students constituted the control group.

For the experimental group, the game-based review system was introduced in addition to traditional review methods, whereas for the control group, students reviewed material along with a teacher's oral explanation, as in traditional methods. The average student age was 10. Both classes were taught by the same teacher, who had 10 years of experience teaching social studies.

3.2 The Design of Game

The game is entitled "Rescue the Princess." The prince must advance through multiple levels, duel with the boss, and ultimately save the princess. At the beginning of each level, the plot and procedure are explained (Fig. 1). The engaging game scenarios attract students to the game. Students can advance at personalized speeds to complete the game.

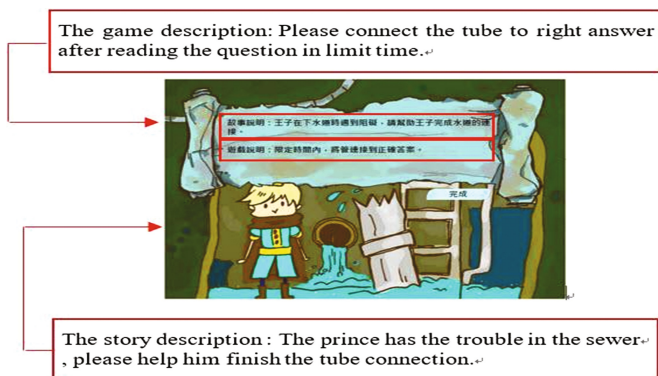


Fig. 1. The description of the story and explanation the operation at the beginning of level.

The game contains five levels (Fig. 2), all of which were designed to enhance student memory of what was previously taught. Level 1 requires students to read questions and guide the prince to the answer using a pipeline jigsaw within a limited time. Students can organize their own routes to the destination to increase their ability of diverse thinking. Most Level 2 questions are multiple choice. Students must flip cards to answer, and, numerous cards appear below the questions. Students are required to find and flip two identical cards to derive the answer. The answer card is then automatically transferred to the answer box. This level was designed so that students can experience a variety of stimulating factors and improve both their memory and their ability to distinguish. For Level 3, guests are invited to join. Students must pass the answers on top to the guests to solve their issues. Students must read and answer the questions quickly, thereby becoming excited and engaged in the game. Level 4 contains multiple choice questions. After reading the questions, students must shoot balloons with answers inside. For Level 5, the prince must answer questions accurately to challenge the boss. Otherwise, the prince is attacked by the boss.

The game emphasizes the function of reviewing. When students provide incorrect answers, the game displays additional content related to the question. Students can then circle the relevant answers manually, and then restart the game. The manual circle function enables students to gain memory points through reading and manual operations in the reviewing part.

Regarding learning content, level records, operations, and other relevant student records are recorded in a system log in detail.



Fig. 2. The five levels in the game based review system.

3.3 Procedure

This study utilized course material from a fourth-grade history class. The content included two lessons. Games were designed to correspond to each lesson. Figure 3

shows the experiment content. The experimental and control groups received a pretest questionnaire regarding interest and attitude before the experiment began. Prior to being taught the first unit, the game and relevant operations were explained to the experimental group. The teacher then taught Lesson 1 for 100 min. Within 20 min following the lecture, the control group reviewed the lesson with their teacher; in the same period, the game was introduced to the experimental group. The game had three levels. A test on Lesson 1 began after the review. The steps for Lesson 2 were similar to those for Lesson 1; however, the models of two of the three levels were different. A posttest was taken in the 9th week. Finally, a posttest questionnaire about interest and attitude of learning was distributed. Throughout the experiment, the teacher and the programmer were available to the students to provide timely assistance.

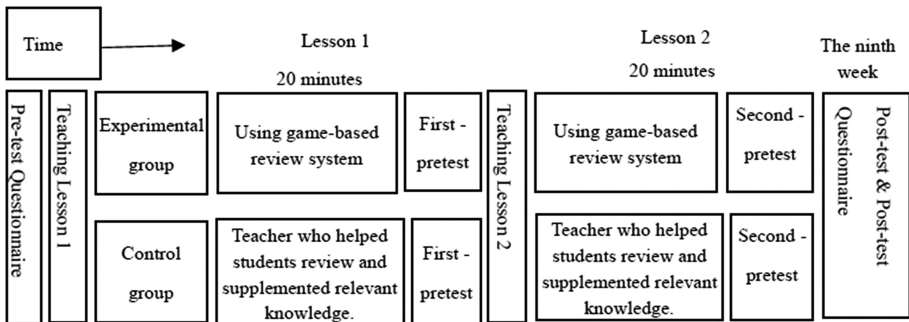


Fig. 3. Experiment procedure

3.4 Assessment Tools

The research tool was self-designed game review software. It included a game map that allowed learners to progress step-by-step through the learning course modules. Every level had questions related to each lesson. Supplementary questions were incorporated into the levels to enable students to succeed and review areas they have made mistakes in or were unfamiliar with, thus improving their knowledge of the relevant topic. In addition, the experiment contained three tests: first pretest, second pretest, and posttest. All questions were in multiple choice. The game employed choice questions, matching questions, and linking games. All test questions were designed by the aforementioned history teacher.

The questionnaires on learning attitude and learning interest were quoted from Hwang and Chang (2011) [13]. The questionnaire regarding learning attitude contained eight questions, which were about learning attitude and the significance of learning. The questionnaire regarding learning interest contained four questions, focusing on the student interest in the course material.

4 Result

4.1 Learning Achievement

Study effectiveness analysis was based on the first pretest, second pretest, and posttest scores of the experimental and the control groups. All the three tests entailed employing an independent sample t test to calculate the learning differences between the two groups; statistical significance was set at $p < 0.05$. Table 2 displays the scores of the experimental group ($M = 61.43$, $SD = 28.202$) and the control group ($M = 64.78$, $SD = 27.858$). The average score of the experimental group was lower than that of the control group. Moreover, no significant difference was observed between the two groups ($p = 0.374$). This is because students in the experimental group exhibited different behavior patterns and experienced a greater change in the learning model than the control group did. Hence, the results of the first pretest for the experimental group were low. In the second pretest and posttest, although the scores of the two classes were similar, the average scores of the two tests of the experimental group were higher than those of the control group. This indicates that the introduction of the tablet game significantly improved student academic performance. According to the posttest results, students provided with the game-based review system had retained more information than the control group did.

Table 2. Learning achievement result

	Group	N	Mean	Std. Deviation	t	p
First-pretest	Experimental	21	61.43	28.202	-.397	.374
	Control	23	64.78	27.858		
Second-pretest	Experimental	21	60.48	23.817	.329	.372
	Control	23	58.26	28.492		
Post-test	Experimental	21	75.62	17.107	.562	.2885
	Control	23	72.78	16.368		

4.2 Learning Attitude

Prior to the experiment, the two classes showed no significant differences regarding learning attitude. Before the introduction of the game-based learning approach, the two classes had the same attitude toward the study of social studies ($p = 0.202$, $t = -3.97$). Table 3 indicates that after the experiment, the two classes' learning attitudes had

Table 3. Learning attitude result

	Group	N	Mean	Std. Deviation	t	p
Pre-test	Experimental	21	4.97	0.689	0.842	0.202
	Control	23	4.74	1.035		
Post-test	Experimental	21	4.95	0.736	1.996	0.026 ^a
	Control	23	4.46	0.88		

^a $p < 0.05$

significant differences ($p = 0.026$). In the research environment, because of less employment of relevant devices in the teaching environment, the game and tablet were used in teaching. The questionnaire on learning attitude contained eight questions, and its Cronbach's α value was 0.817.

4.3 Learning Interest

In the pretest, the experimental group ($M = 4.6$, $SD = 1.065$) and the control group ($M = 4.13$, $SD = 0.859$) did not exhibit any significant differences in terms of learning interest levels ($p = 0.54$). However, subsequent interest level assessments revealed that the scores of the experimental group ($M = 4.75$, $SD = 1.092$) and those of the control group ($M = 4.05$, $SD = 1.081$) were significantly different ($p = 0.02$). The application of the proposed game-based review software enhanced student fondness for the courses and intensified their interest. The questionnaire regarding learning interest contained four questions, and its Cronbach's α value was 0.875 (Table 4).

Table 4. Learning interest result

	Group	N	Mean	Std. Deviation	<i>t</i>	<i>p</i>
Pre-test	Experimental	21	4.6	1.065	1.641	0.54
	Control	23	4.13	0.859		
Post-test	Experimental	21	4.75	1.092	2.12	0.02 ^a
	Control	23	4.05	1.081		

^a $p < 0.05$

5 Conclusion

Although the scores of the first pretest, second pretest, and posttest of the experimental group and control group did not reach significance, the average scores of the first pretest and second pretest of the experimental group were higher than those of the control group because of the introduction of the review system. The system combined game levels with questions and provided additional review functions following incorrect answers. Students could read content related to the incorrectly answered question and manually circle keywords to retain the information. Additionally, the introduction of mobile devices and gamed-based review software improved student learning attitude. As the students advanced in the game, they expressed desire to learn the content of the level to pass it, which enhanced their awareness of goals and initiative and provided them with a unique learning experience. The multimedia features of the implemented game-based review software improved learning interest. Students were happy about what they learned, and their interest and willingness to learn increased. Students exhibited enthusiasm and devoted time to the course, thereby enhancing their learning effectiveness and attitude. Future studies can examine behavior and motivation to conduct more in-depth research and discussions to serve as reference for scholars in the relevant fields.

Acknowledgements. This work was supported in part by the Ministry of Science and Technology (MOST), Taiwan, ROC, under Grant MOST 103-2511-S-224-004-MY3, MOST 104-2511-S-224-003-MY3, and MOST 105-2628-S-224-001-MY3.

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